

Network Architectures and Protocols

List of Exam Questions

Date: 17th May 2018

- 1) List ten important milestones of the Internet technology evolution.
- 2) Give four classification criteria of the computer network and give examples of each category.
- 3) Explain following mechanisms: signal coding, modulation, multiplexing, demultiplexing, transmission rate, modulation rate (text and figure).
- 4) Characterize the analog and digital transmission (text and figures).
- 5) List three switching modes and characterize them.
- 6) List four addressing methods and characterize them.
- 7) Compare three layer models of the communication and explain the layer functions (text and figure).
- 8) List five intermediate nodes and characterize them in detail (text and figure).
- 9) Characterize the following network mediums: coaxial cable, fibre optic (text and figures).
- 10) Explain the impact of noise on communication and characterize the SNR term (text and figures).
- 11) Compare unipolar and bipolar coding mechanisms and give two examples of each group (text and figure).
- 12) List four modulation technics and explain their mechanism (text and figure).
- 13) List four topology types and characterize them (text and figure).
- 14) List static and dynamic channel access mechanisms and characterize them.
- 15) Similarities and differences of the ALOHA and Slotted ALOHA communication mechanisms (text and figure).
- 16) Describe Code Division Multiple Access mechanism.
- 17) Description of the IEEE 802.3 MAC algorithm (text and figures).
- 18) Description of the IEEE 802.3 MAC frame structure (text and figures).
- 19) Compare different Ethernet technology versions.
- 20) Description of the IEEE 802.5 MAC algorithm (text and figures).
- 21) Description of the IEEE 802.5 MAC frame structure (text and figures).
- 22) Compare the datagram characteristics on L2, L3, L4 and give concrete communication technology example for each level (text and figures).
- 23) Describe IPv4 packet structure and functions of the fields (text and figure).
- 24) Describe IPv6 packet structure and functions of the fields (text and figure).
- 25) Characterize the IPv6 address types with examples (text).
- 26) List routing protocol groups and give examples of them.
- 27) Compare TCP and UDP mechanism (text and figure).
- 28) Explain the role of the physical address, logical address and port identifier.
- 29) Describe the E-mail service (text and figure).
- 30) Describe the FTP and the HTTP mechanisms (text and figures).